

Gate E2 report — run_v1.21_r1 (Paper IV v1.21)

Verdict: **INCONCLUSIVE**. Evidence: 20_EVIDENCE/E2/.

E2 — Mathematics revalidation (v1.21 against v1.2 findings)

Method: the auditor's own unified diff of v1.2 $\rightarrow$ v1.21 (auditor_diff_en.diff, auditor_diff_es.diff), pen-and-paper rederivation of every added passage, and exact checks of the new numerics (20_EVIDENCE/E5/e5_v121_numerics.py, 21 checks, all PASS, all negative controls rejected). The author response (RESPONSE_TO_AUDIT_v1.21.md) was read only after these conclusions were written. No Lean was run (mandate §2).

Theorems unchanged

No theorem statement, hypothesis, constant, equation tag or Lean code block changed (diff). The only formula edits are the declared descriptive subscripts: $N_{\text{lej}}/N_{\text{lejano}} \rightarrow N_{\text{far}}$, $e_{\text{cruz}} \rightarrow e_{\text{cross}}$, $x^{\text{limpio}} \rightarrow x^{\text{clean}}$, $E_{\text{cub}} \rightarrow E_{\text{covered}}$. These are semantic no-ops (N_{far} already named the same far-branch threshold in (C.4)).

A.2 items — rederivation and verdicts

Item	v1.21 location	Rederivation performed	Verdict	Explanation
A2-1	§5.1, new sentence before "The retained columns..."	Yes. X-columns miss $\geq a/4$ of $\leq 129a/64$ old exterior vertices $\Rightarrow \leq 113a/64$; non-Y exterior degree $< 7a/4 \leq 113a/64$; $+ X $	ACCEPTABLE_SUMMARY	X-23 closed.
A2-2	C.2 (3.10), unchanged	As in run_v1.2_r1	ACCEPTABLE_SUMMARY	Unchanged; standard bookkeeping.
A2-3	C.2, four new paragraphs after (3.13)	Yes, complete. (i) Load: surviving copies through a good root $\leq (1+u)A_{\{H,q\}} = (1+u)V_H/d_q$, weight $\psi_H/((1+u)V_H) \Rightarrow$ contribution $\leq \psi_H/d_q$; with $\Sigma_{\{H \text{ serving } q\}} \psi_H \leq d_q$ the load is ≤ 1 . (ii) Retention: $B_q \leq 118t^2/(u^2a_3^2)$ (K_4 : $238t^2/(u^2a_4^2)$); $\leq t$ (resp. t^2) copies per root; 3 (resp. 6) pairs $\Rightarrow \leq 338t^3/(u^2a_3^2)$, $1388t^4/(u^2a_4^2)$ deleted; with $V_H \geq a_3t^3$ (a_4t^4) the deleted fraction is $\leq v$ under $338 \leq vu^2a_3^3$, $1388 \leq vu^2a_4^3$. (iii) Schedule: $d = u = v = a$, $\delta = a^{21}/2208$, $a_3 \geq a^3/2$, $a_4 \geq a^6/2$, $1388 = a^{21}/16 \leq a^{21}/8$ (E5 C2-1... C2-3). (iv) Weight retained $\geq (1-v)/(1+u)$.	ACCEPTABLE_SUMMARY	X-01 resolved. Residual OBSERVATION R-01: the second-moment inputs " $B_q u^2a_3^2 \leq 118t^2$ ", " $B_q u^2a_4^2 \leq 238t^2$ " are cited from the rooted second-moment (counting-lemma) estimates, not derived; they are standard consequences of δ -regularity. Wording note R-02: "H serving q" must mean every active profile containing the pair q (the load bound needs all profiles through q); with the earlier sentence "every active profile serves a pair of classes" the reader could take it as one pair per profile. Checked in the frozen source: <code>PatternTransfer.servingT</code> is exactly the set of patterns containing both parts of the pair, so the argument is correct under that reading.
A2-4	C.3, new schedule definitions and numerical chain	Yes for the schedule $\rightarrow$ selector $\rightarrow$ three integers: $\beta_0 \geq 2^{-66}$, $L_T \leq 46$, $M_T \leq 5153$, $\gamma_T \geq 2^{-59483}$, $T_T \leq 2^{\{a_0+13\}}$, $2G_T \leq 2^{\{4a_0+5\}}$, $(2G_T)^{\{T_T\}} \leq 2^Y$, $\eta_T \geq 2^{\{-(Y+69)\}}$, $\ell_T \geq 2^{-12749}$, $D_0 \leq 2^{\{a_0+146\}}$, $c_0 \geq 2^{\{-(Y+3a_0+220)\}}$, $Z = 191424$, $U = Y+191432$, $G_3 \leq 2^{\{U+67\}}$ (E5 C3-1...C3-16). Not possible for the last step: "The regularity recurrence, iterated h times ..., dominates these three quantities after just its first two iterations. Absorbing the five terms of (C.1) then gives $N_{\text{far}}(\eta_0) \leq T(h+7)$." The regularity bound B (C.3 l.1646) and its recurrence are never defined in the manuscript, and A.2 (l.1355) refers to "the recurrence in §6.3", but §6.3 defines only T and h.	REQUIRES_EXPANSION (MINOR impact, narrowed)	X-05 partially resolved. Exact remaining step: state the regularity bound used (the recurrence whose h-fold iterate bounds B at parameters $(\delta/8, k_0)$; $h = 4/(\delta/8)^5$ in the displayed factorization), prove the two-iteration domination of $G_1...G_3$, and show the absorption of the five (C.1) terms into $T(h+7)$. Affects only the explicit values $T(h+7)$, $T(h+8)$ (§6.3) and the $c_{\text{far}}/h_{\text{iter}}$ part of $P(s)$ (6.28), not the existence of thresholds or any theorem. The author's response also says this comparison "remains symbolic".
A2-5	E.1, three new paragraphs (normalization counts) and σ , τ , D, L	Yes, complete: $\varepsilon n^2 = \lambda^2/10^{41}$, $n\lambda = \rho^2$; $ A A_0 \leq \lambda/10^{31}$; $ A_0 $ window; $e(R_0) \leq 2\rho^2/10^{31}$; $ Y \leq 4\rho/10^{27}$; $Y_c/Y_r/W$ partition; $ c-n/3 \leq \rho/10^{20}$; light-vertex missing $\leq \rho/10^4 + 2\lambda/10^{41} < \rho/200$; $\sigma \leq \rho/200 + 12\rho/10^{27}$; $\tau \leq \rho/40 + 12\rho/10^{27}$; $D \leq (\rho/10^{13})^2$; $L = \lfloor \sqrt{D} \rfloor + 1$ (E5 E1-1).	ACCEPTABLE_SUMMARY	X-02 resolved. The localization input (clique A with ε -bounds and minimum degree) is named as produced by fixed-defect localization (E33/E34, [22]); that is a separately attributed component, not part of A2-5.

A2-6	E.1, new derivation of (E.2d)	Yes, complete: choice of a_w ($w \in W$ misses $> \rho/200$ in A_0), Z = common light neighbours of U_0 , three-case elimination with charges $s-v$, $\omega+s$, $(\omega+s)x$, each incidence charged once; $e(Z, X') \leq (s-v) Z + 2(\omega+s)x$; restoration $ Z \leq 2v\rho/40 + x\lambda/10^{10}$; error $\leq n/10 + 4n/10^4 + 4\rho/10^{10} + 4\rho/10^{50} \leq n/9$ (E5 E2d-1, negative control $n/12$ rejected).	ACCEPTABLE_SUMMARY	X-03 resolved.
A2-7	D.1, unchanged	As before	ACCEPTABLE_SUMMARY	Unchanged.

Other E2 items

- The added Figure 3 citation and the audit-status paragraphs carry no mathematics.
- X-24 (optional constant tightenings) not adopted: acceptable.

E2 verdict

INCONCLUSIVE, narrowly: six of seven A.2 rows are now ACCEPTABLE_SUMMARY, and X-01, X-02, X-03, X-23 are resolved. **A2-4 still REQUIRES_EXPANSION** at one exact step (regularity recurrence and final tower comparison in C.3, plus the dangling "recurrence in §6.3" reference). No mathematical error was found.

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